

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 01
Course Title : Structured Programming Language Lab
Course Code : CSE-101
Experiment Name: If-Else

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Lab Report (*if-else*)

Problem Name:

1. Write a C program to input temperature in Centigrade and convert to Fahrenheit.

Source Code:

```
#include <stdio.h>

int main(){
    float c, f;

    printf("Write in centigrade\n");

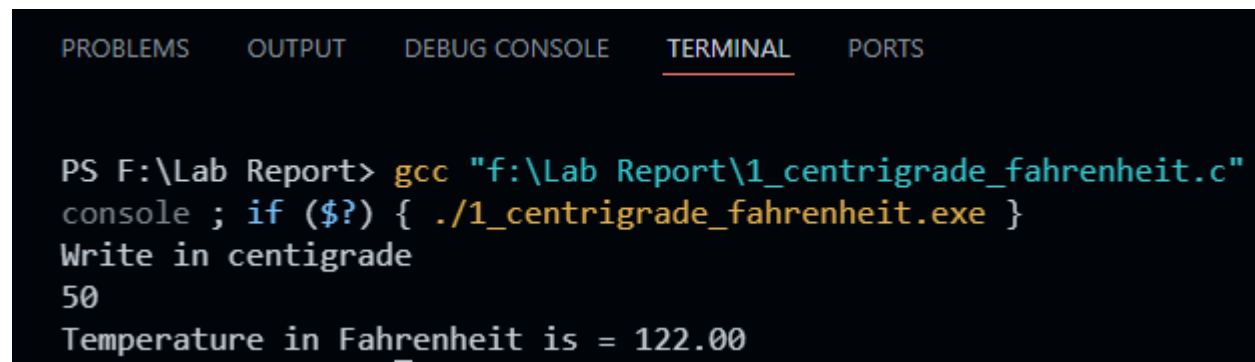
    scanf("%f", &c);

    f= (c*9/5)+32;

    printf("Temperature in Fahrenheit is = %.2f ",f);

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\1_centrigrade_fahrenheit.c"
console ; if ($?) { ./1_centrigrade_fahrenheit.exe }
Write in centigrade
50
Temperature in Fahrenheit is = 122.00
```

Problem Name:

2. Write a C program to input radius of a circle from user and find diameter, circumference and area of the circle.

Source Code:

```
#include <stdio.h>

#define PI 3.1416
```

```

int main()
{
    float r, d, c, a;

    printf("What is Radius?\n");

    scanf("%f", &r);

    d = 2 * r;

    c = 2 * PI * r;

    a = PI * r * r;

    printf("The Circle's diameter is = %.2f , circumference is = %.2f , area is = %.2f\n", d, c,
a);

    return 0;
}

```

Output:



```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
PS F:\Lab Report> gcc "f:\Lab Report\2_circle_dca.c" -o 2_circle_dca.exe -mconsole
circle_dca.exe }
What is Radius?
4
The Circle's diameter is = 8.00 , circumference is = 25.13 , area is = 50.27

```

Problem Name:

3. Write a C program to input any two numbers from user and swap values of both numbers using third variable, bitwise operator and arithmetic operators.

Source Code:

```

// Swap using third variable-

#include <stdio.h>

int main()
{
    int a, b, temp;

    printf("Enter two numbers: ");

```

```
scanf("%d %d", &a, &b);
printf("Original values: a = %d, b = %d\n", a, b);
temp = a;
a = b;
b = temp;
printf("After swapping using third variable: a = %d, b = %d\n", a, b);
return 0;
}
```

// Swap using bitwise XOR-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    printf("Original values: a = %d, b = %d\n", a, b);
```

```
    a = a ^ b;
```

```
    b = a ^ b;
```

```
    a = a ^ b;
```

```
    printf("After swapping using bitwise XOR: a = %d, b = %d\n", a, b);
```

```
    return 0;
```

```
}
```

//Swap using arithmetic operators-

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    printf("Enter two numbers: ");
```

```
scanf("%d %d", &a, &b);

printf("Original values: a = %d, b = %d\n", a, b);

a = a + b;

b = a - b;

a = a - b;

printf("After swapping using arithmetic operators: a = %d, b = %d\n", a, b);

return 0;

}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\3.1_swap_numbers.c"
{ ./3.1_swap_numbers.exe }
Enter two numbers: 4
3
Original values: a = 4, b = 3
After swapping using third variable: a = 3, b = 4
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\3.2_swap_numbers.c"
{ ./3.2_swap_numbers.exe }
Enter two numbers: 3
4
Original values: a = 3, b = 4
After swapping using bitwise XOR: a = 4, b = 3
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\3.3_swap_numbers.c" -o
{ ./3.3_swap_numbers.exe }
Enter two numbers: 10
12
Original values: a = 10, b = 12
After swapping using arithmetic operators: a = 12, b = 10
```

Problem Name:

4. Write a C program to check maximum between two numbers using ladder if else or nested if.

Source Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c;
```

```
    printf("Write 3 numbers\n");
```

```
    scanf("%d%d%d", &a, &b, &c);
```

```
    if (a >= b && a >= c)
```

```
    {
```

```
        printf("a =%d is the greatest\n", a);
```

```
    }
```

```
    else if(b >= a && b >= c)
```

```
    {
```

```
        printf("b= %d is is the greatest\n", b);
```

```
    }
```

```
    else
```

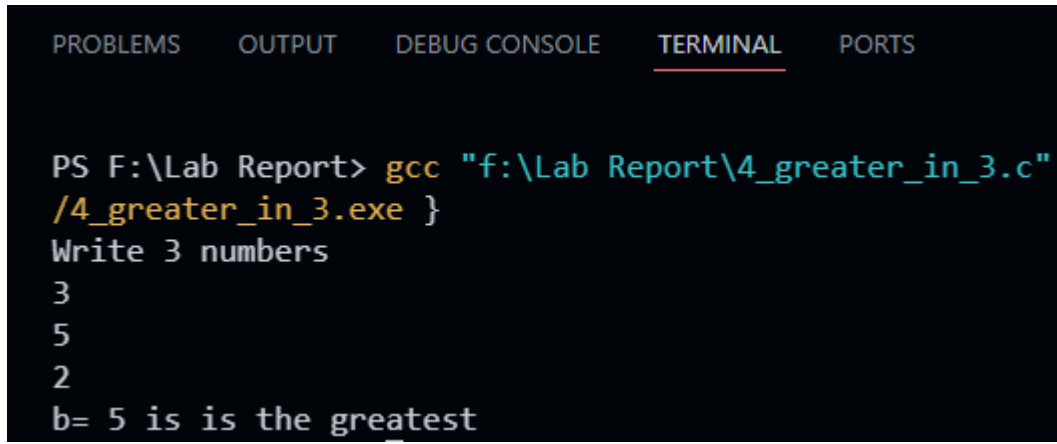
```

{
    printf("c=%d is the greatest\n", c);
}

return 0;
}

```

Output:



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\4_greater_in_3.c"
/4_greater_in_3.exe }
Write 3 numbers
3
5
2
b= 5 is is the greatest

```

Problem Name:

5. Write a C program to check a given year is leap year or not.

Source Code:

```

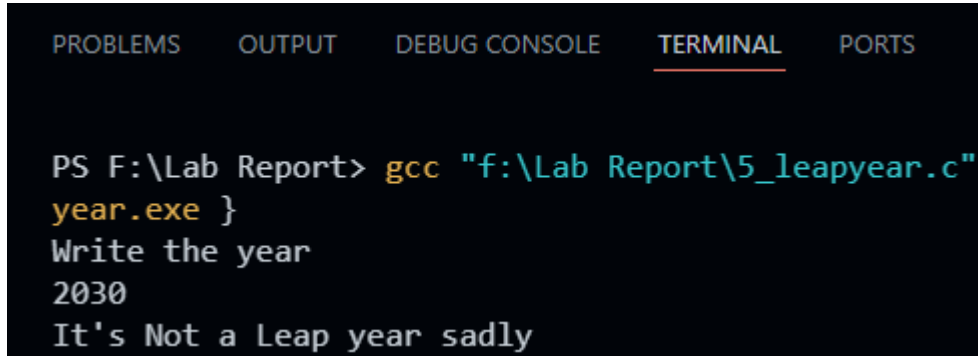
#include <stdio.h>

int main()
{
    int y;
    printf("Write the year\n");
    scanf("%d", &y);
    if ((y % 400 == 0) || (y % 4 == 0 && y % 100 != 0))
        printf("It's a Leap year!\n");
    else
        printf("It's Not a Leap year sadly\n");
}

```

```
    return 0;
}
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS F:\Lab Report> gcc "f:\Lab Report\5_leapyear.c"
year.exe }
Write the year
2030
It's Not a Leap year sadly
```

Problem Name:

6. Write a C program to input a character from user and check whether given character is alphabet, uppercase, lowercase, digit or special character.

Source Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char ch;
```

```
    printf("Enter a character\n");
```

```
    scanf("%c", &ch);
```

```
    if (ch >= 'A' && ch <= 'Z')
```

```
        printf("It's an uppercase alphabet\n");
```

```
    else if (ch >= 'a' && ch <= 'z')
```

```
        printf("It's a lowercase alphabet\n");
```

```
    else if (ch >= '0' && ch <= '9')
```

```
        printf("It's a digit\n");
```

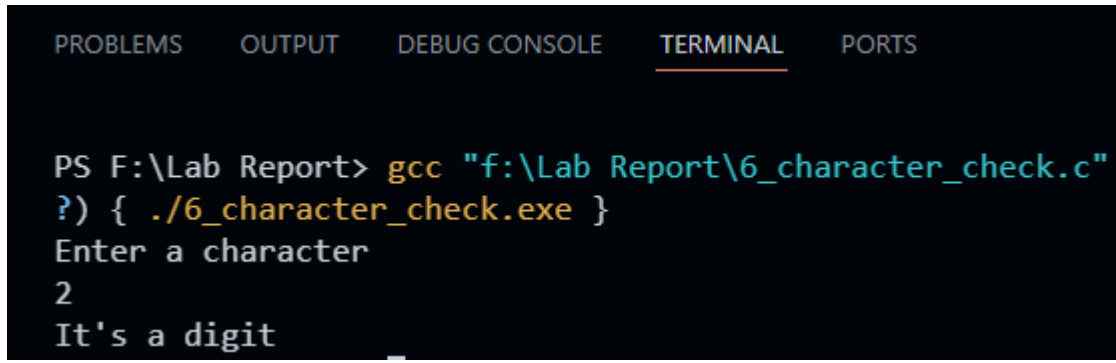
```
    else
```

```
        printf("Special Charracter\n");
```



```
    return 0;
}
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS F:\Lab Report> gcc "f:\Lab Report\6_character_check.c"
?) { ./6_character_check.exe }
Enter a character
2
It's a digit
```

Problem Name:

7. Write a C program to check whether an alphabet is vowel or consonant.

Source Code:

```
#include <stdio.h>
```

```
int main(){
    char ch;

    printf("Write an Alphabet:\n");
    scanf("%c", &ch);

    if (ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U' || ch=='a' || ch=='e' || ch=='i' || ch=='o' ||
ch=='u')
        printf("It's a vowel\n");
    else
        printf("It's a consonant\n");

    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS F:\Lab Report> gcc "f:\Lab Report\7_vowel_consonant.c"
$?) { ./7_vowel_consonant.exe }
Write an Alphabet:
h
It's a consonant
```

Problem Name:

8. Write a C program to calculate grade of student, If(i) number ≥ 80 : Grade A+, (ii) number ≥ 70 : Grade A, (iii) number between 60 to 70: Grade A-, (ii) number between 50 to 60: Grade B, (ii) number ≥ 40 : Grade C and number < 40 : Grade F.

Source Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int g;
```

```
    printf("Write the number you got-\n");
```

```
    scanf("%d", &g);
```

```
    if (g  $\geq$  80)
```

```
        printf("Grade A+");
```

```
    else if (g  $\geq$  70)
```

```
        printf("Grade A");
```

```
    else if (g  $\geq$  60)
```

```
        printf("Grade A-");
```

```
    else if (g  $\geq$  50)
```

```
        printf("Grade B");
```

```
    else if (g  $\geq$  40)
```

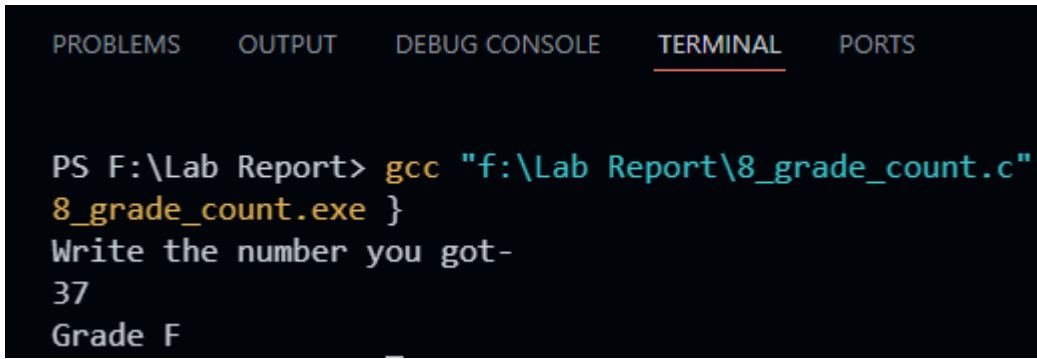
```
        printf("Grade C");
```

```
else

    printf("Grade F\n");

return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\8_grade_count.c"
8_grade_count.exe }
Write the number you got-
37
Grade F
```

Problem Name:

9. Write a C program to input week number and print day of week name using switch case.

Source Code:

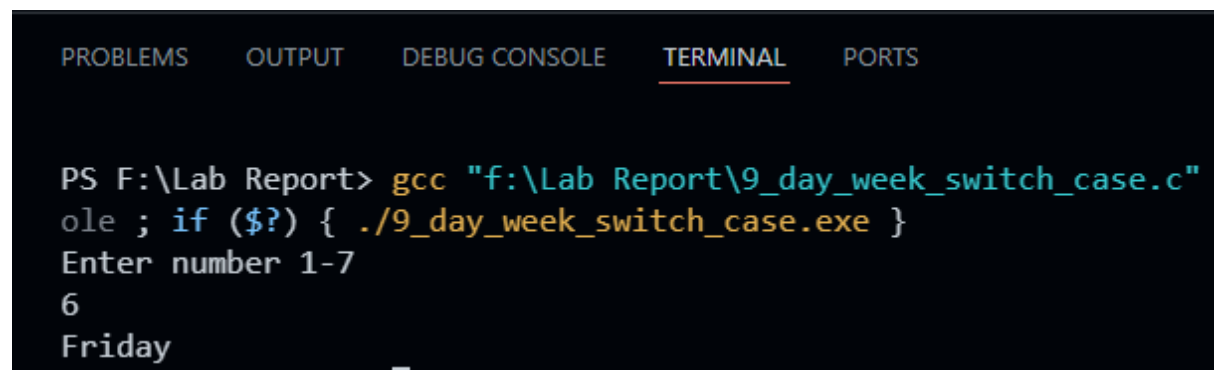
```
#include <stdio.h>

int main()
{
    int day;

    printf("Enter number 1-7\n");
    scanf("%d", &day);
    switch (day)
    {
        case 1:
            printf("Sunday\n");
            break;
        case 2:
            printf("Monday\n");
```

```
        break;
case 3:
    printf("Tuesday\n");
    break;
case 4:
    printf("Wednesday\n");
    break;
case 5:
    printf("Thursday\n");
    break;
case 6:
    printf("Friday\n");
    break;
case 7:
    printf("Saturday\n");
    break;
default:
    printf("Invalid input, Enter number beteen 1-7 \n");
}
return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\9_day_week_switch_case.c"
ole ; if ($?) { ./9_day_week_switch_case.exe }
Enter number 1-7
6
Friday
```

Problem Name:

10. Write a C program to input a number and check whether given number is odd or even.

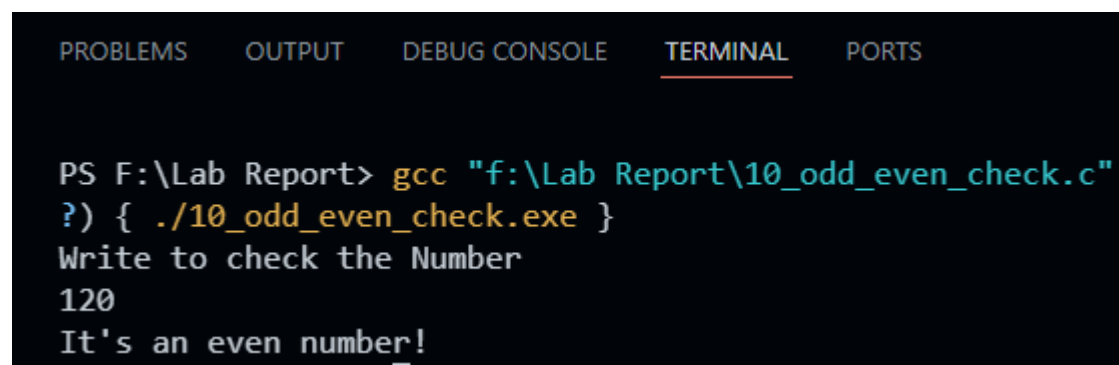
Source Code:

```
#include <stdio.h>

int main(){
    int num;

    printf("Write to check the Number\n");
    scanf("%d", &num);
    if (num%2==0)
    {
        printf("It's an even number!\n");
    }
    else
    {
        printf("It's an odd number!\n");
    }
    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\10_odd_even_check.c"
?) { ./10_odd_even_check.exe }
Write to check the Number
120
It's an even number!
```